CSI/ELG/SEG 2911 (Winter 2026)

Professional Practice in Engineering and Computing

Guidelines and Syllabus

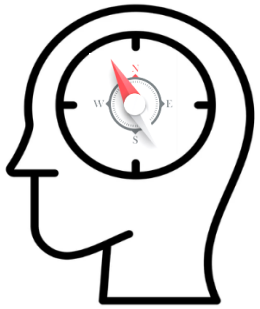
Lectures: Thursdays (7:00 pm – 10:00 pm)
Room: CRX C240

uOttawa

Faculté de génie
Faculty of Engineering

Course Instructor: Riadh Habash (rhabash@uottawa.ca)
Corrector (Cases 1 and 3): Niloofar Jazaeri (njaza024@uottawa.ca)
Corrector (Cases 2 and 4): Phavay Malhotra (bmalh018@uottawa.ca)
Corrector (Project): Jerry Wang (jwang623@uottawa.ca)
Corrector (Project): Romina Etezadi (retez068@uottawa.ca)

The Big Questions: How do ethical principles—such as public safety, professional competence, truthful communication, responsibility, fair competition, lifelong learning, and equitable treatment—guide decision-making, innovation, public welfare, and daily engineering choices? Why does this moral compass matter to engineers and computing professionals, as well as the industries they serve and society?



This course does not have any additional costs.

Guidelines

There is no textbook required for this course, and there are no extra costs

Lectures: Thursday from 7:00 pm to 10:00 pm at CRX C240
DGDs: Tuesday (8:30 am - 9:50 am; 2:30 pm - 3:50 pm; Friday 10:00 am - 11:30 am).
All course information, materials, and announcements are posted on **Brightspace**.
The **Course Content** provides detailed information on week-by-week **learning activities**.
All course assignments are posted under the **Assignment** tab.
All phones and other electronic communication devices must be turned to SILENT during lectures.
Late assignment submissions are subject to a **20%** deduction per day.
The Concept Knowledge (Two Tests) is an individual task.
Cases and Projects are group tasks (one; two; or three students). Register your group on the Brightspace **Groups** tab**.**

Objectives

Stimulating ethical professional practice imagination; **developing** ethical knowledge; **identifying** ethical problems; **analyzing** key ethical concepts and principles; **mastering** relevant ethical judgments and willpower; **dealing** with ambiguous problems; and **improving** the responsibility of the educated.

Upon completion, learners should be able to:

Recognize and analyze problems related to the interdisciplinary domain of ethical professional practice.

Enhance awareness of ethical situations and understand issues related to people, society, nature, and technology.

Know how to avoid and solve ethical problems, through practice, involving honesty, fairness, well-being, environmental protection, and war.

Improve professional skills, including personal, interpersonal, thinking, and technical.

Develop awareness of global responsibility and social dimensions in matters related to sustainability, well-being, and intelligence.

Recognize the need for life-long learning to stay informed about applicable technical, ethical, legal, equality, and equity practices.

Canadian Engineering Accreditation Attributes (CEAB)

Graduate Attributes

Qualities, Skills, and Understandings

- Knowledge Base
- Problem Analysis
- Investigation
- Design
- Use of Engineering Tools
- Individual and Teamwork
- Communication Skills
- Professionalism
- Impact of Engineering on Society and Environment
- Ethics and Equity
- Economics and Project Management
- Life-long Learning

Syllabus

Concept Knowledge
Professionalism, Regulations, Ethics, Law & IP, Design Safety, Professional Skills, Leadership, Entrepreneurship Mindset, Sustainability, Project Management, Ethics of AI

Teamwork

Case Research Experience
Equity, Diversity, Inclusion Entrepreneurship, Ethics of Care, AI & Sustainable Development*

Project Research Experience
Design Thinking, Proof of Concept & Value, Professional Skills, Digital Literacy

Experience-based Learning

Assessment: Tests (30 marks) + Cases (40 marks) + Project (30 marks)

Case 1 **(10 marks)**

Case 2 **(10 marks)**

Case 3 **(10 marks)**

Case 4 **(10 marks)**

Test 1 **(15 marks)**

Test 2 **(15 marks)**

Project Proposal **(10 marks)**

Project Presentation **(10 marks)**

Project Article **(10 marks)**

*https://sdgs.un.org/goals

Tentative Schedule of Tasks

January 15	Professionalism and Regulations (Module 1)
January 22	Ethical Practice (Module 2)
January 29	Legal System and Intellectual Properties (Module 3). **Case 1 Submission**
February 05	Communication Skills, Reflective Practice (Module 4)
February 12	Entrepreneurship Mindset (Module 5); Leadership (Module 6). **Project Proposal Submission**
	Study Break
February 26	**Midterm 1 (MCQs and Virtual); Case 2 Submission**
March 05	Design Safety (Module 7)
March 12	Project Management (Module 8); Sustainable Development Goals. **Case 3 Submission**
March 19	AI Ethical Practice (Module 9)
March 26	**Midterm 2 (MCQs and Virtual). Case 4 Submission**
March 29	**Submission of Project Educational Video (10 marks)**
April 02	Project Video Presentations in the Class
April 09	Project Video Presentations in the Class
April 09	**Submission of Project Paper (10 marks)**

https://janetgregory.ca/implicit-tacit-vs-explicit-knowledge/

Explicit Knowledge-*for-Practice*

Concept knowledge in isolation, living in books and heads!
Codified as data and information in documents and databases. Computing is essential in storage and transfer.

Implicit Knowledge-*in-Practice*

Knowledge insights that operate at a subconscious level!
Can be developed and shared through reflection and translation, such as posters and videos.

Tacit Knowledge-*of-Practice*

Ingrained knowledge and wisdom in integration!
Intuitive, know-how, experiential, unwritten, and tribal knowledge.
Rooted in context, experience, practice, and values.
Skills and experience, but hard to communicate! Requires reflexivity!
Personal knowledge that resides in the mind of the practitioner.
Source of long-term competitive advantage and innovation.
Transfer through socialization and mentoring (learn-by-doing).
Computing may help.

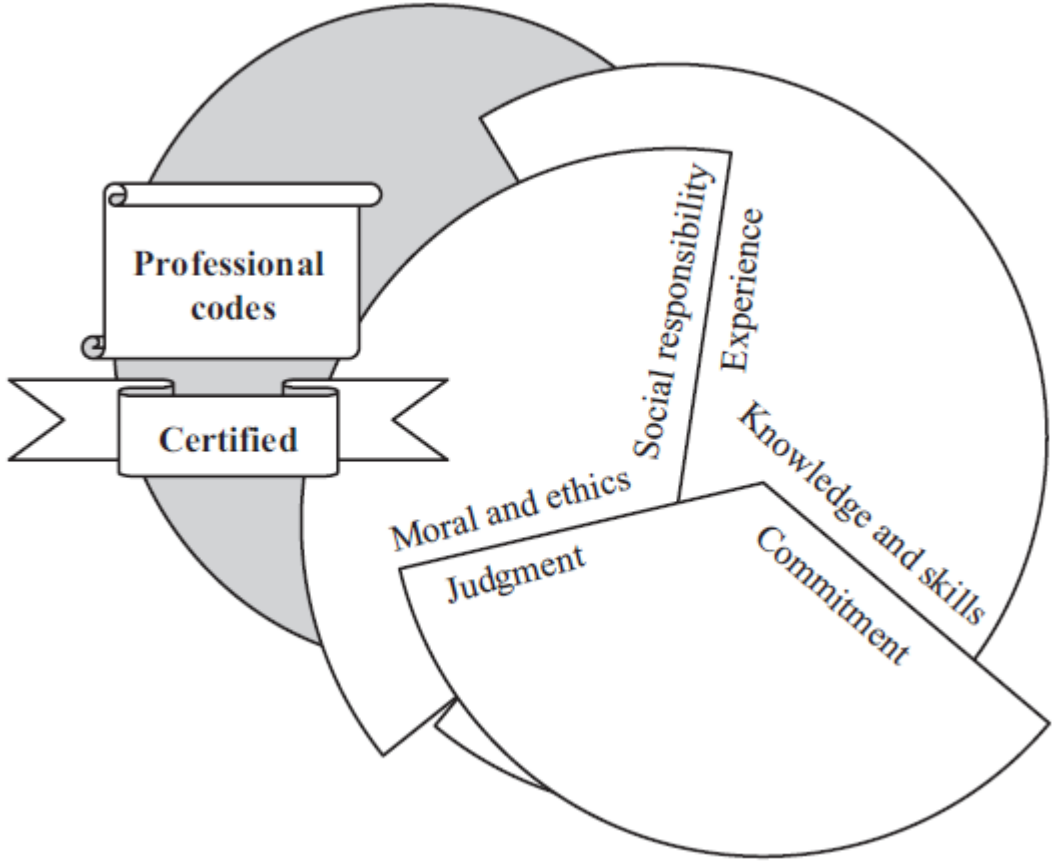
Do you think AI can replicate tacit knowledge?

The highest form of human intelligence is tacit knowledge

Realizing **Professionalism**

It's not the job you do; it's how you do the job. **Anonymous**

Professionalism, in the sense perceived in the quotation above, is more than doing a job; it is a key cluster of social and ethical responsibility, commitment, knowledge and skills, and judgment in the sociologies of work, occupations, professions, and organizations.



Learning and Practicing **Ethics**

How to develop the citizens and not just professionals?

Ethics may be conceptualized by the Greek term **"eudaimonia,"** which may be realized as a combination of well-being, happiness, and flourishing in the tradition of Aristotelian virtue ethics.

This may inspire educators and students to present ethics as aspirational rather than solely preventive. Ethical decision-making should focus on enhancing the well-being of both the public and professionals.

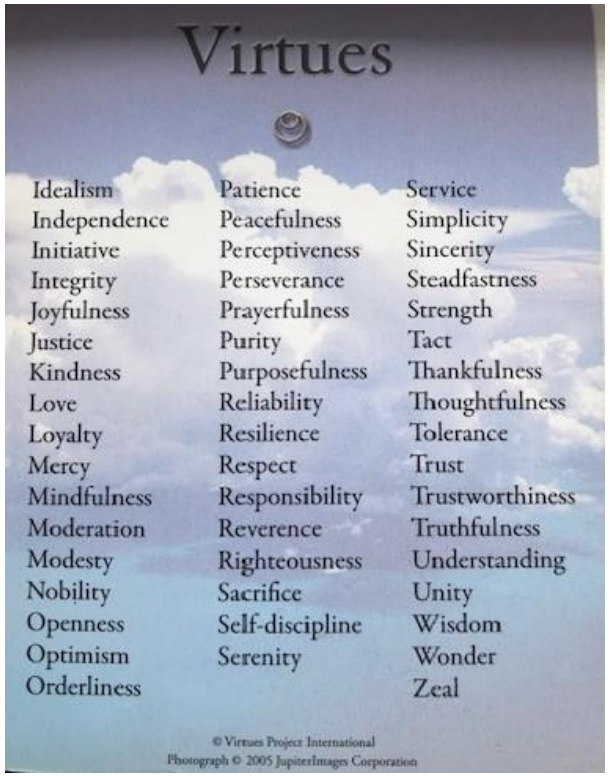
The question is how to develop citizens, not just professionals.

What to Teach in E&C Ethics?

Microethics, Macroethics, Preventive, and Aspirational Ethics

Stimulation of ethical imagination of students to improve ethical judgment and willpower because engineering and computing (E&C) are not just technical or theoretical enterprises.

Analysis of key ethical concepts and principles relevant to E&C such as public health and safety, quality, usefulness, efficiency, cost/risk/benefit analysis, environmental harm, truthfulness, loyalty, etc.



How to Teach E&C Ethics?

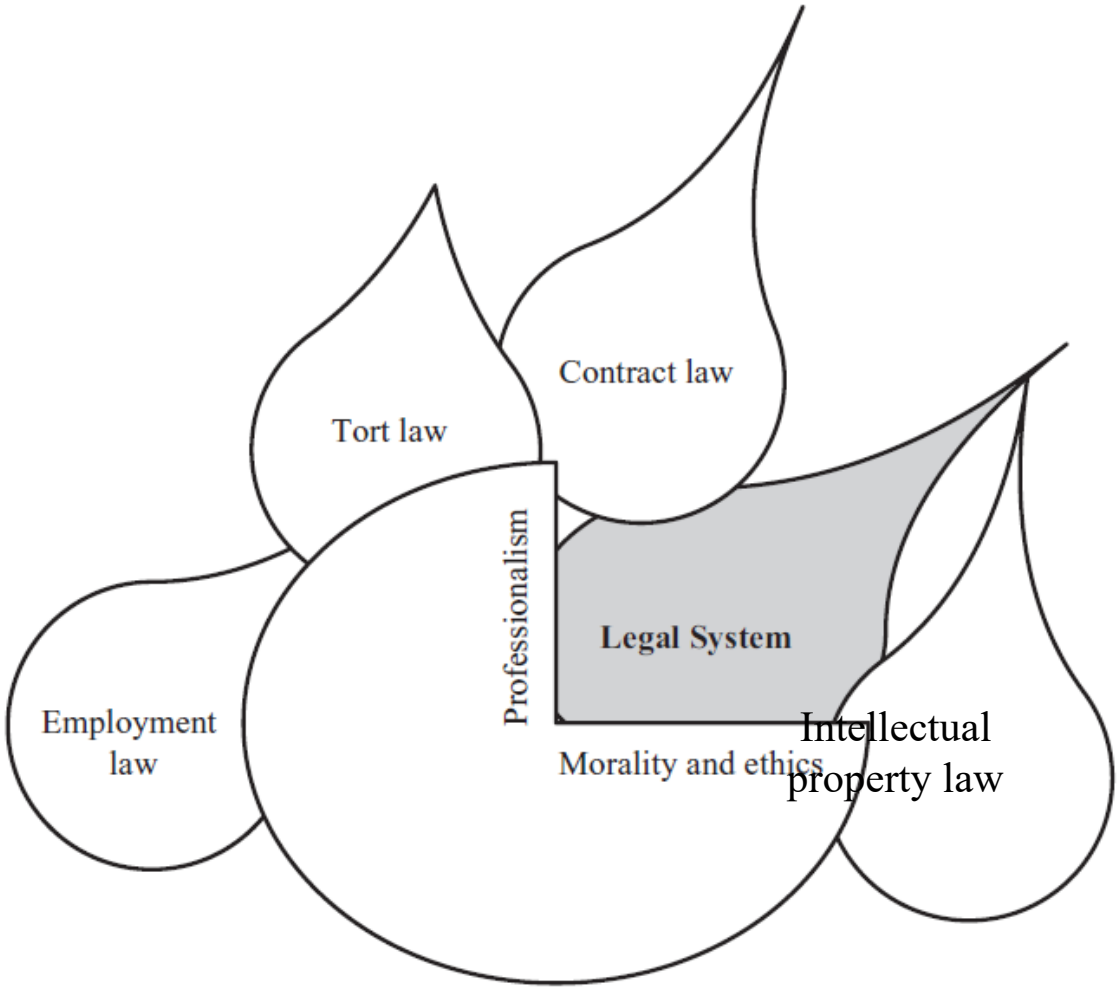
By analyzing aspirational cases, not just disaster cases that make media news, but the kinds of cases that a professional is more likely to face.

By introducing E&C ethics into required courses.

By passing out a code of ethics at the beginning of the term and referring to it often enough during the term so that students get the idea it would be good to read it.

Realizing **Legal System Practice**

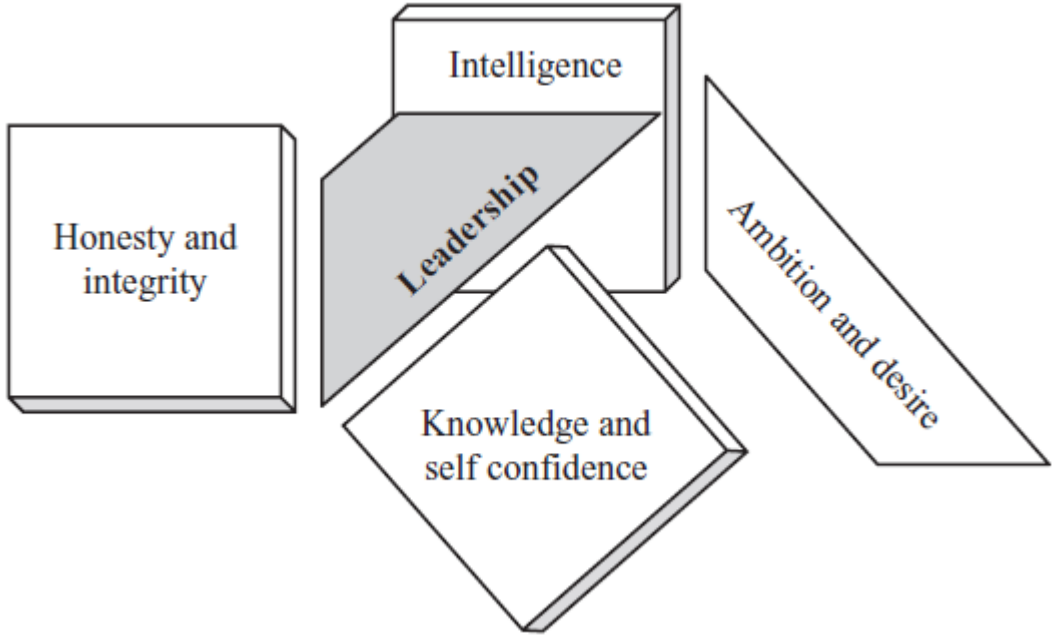
The legal system, in the sense perceived in the quotation, is not just about a set of laws but goes beyond that to embrace morality, professionalism, codes of ethics, discipline, and rules of law.



At the end of the day, you are your own lawmaker. **Bangambiki Habyarimana**

Learning **Leadership Practice**

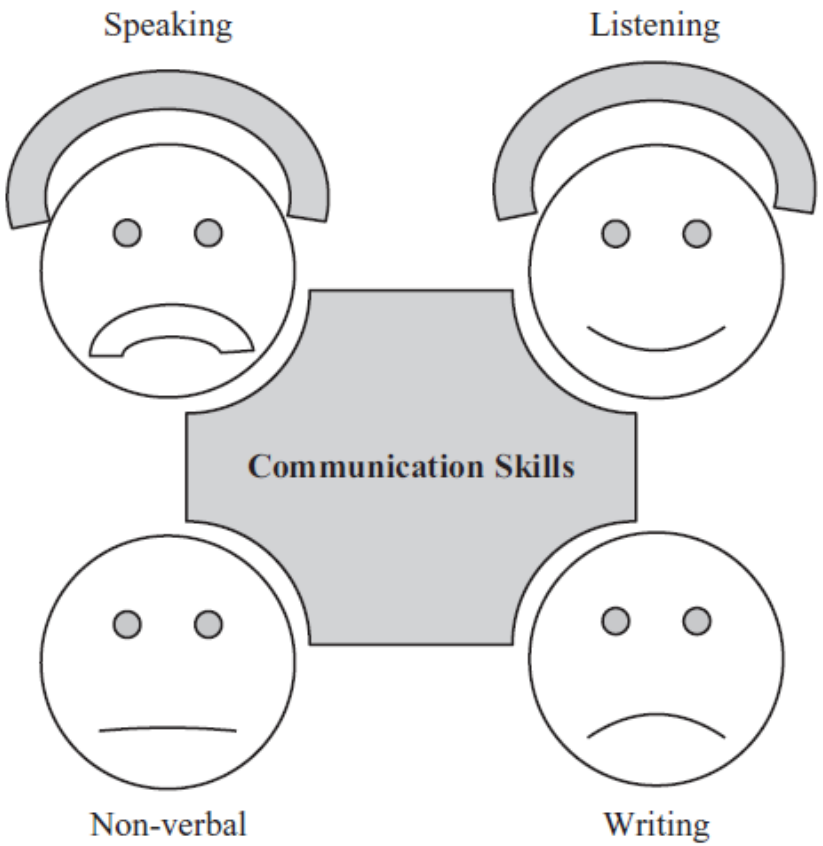
Leadership, in the sense perceived in the quotation, integrates several traits as mutually reinforcing concepts. Leadership can be defined as a collective of ethical actions that influence and direct performance toward the achievement of common goals.



Leadership consists not in degrees of technique but in traits of character. **Lewis H. Lapham**

Learning **Communication Practice**

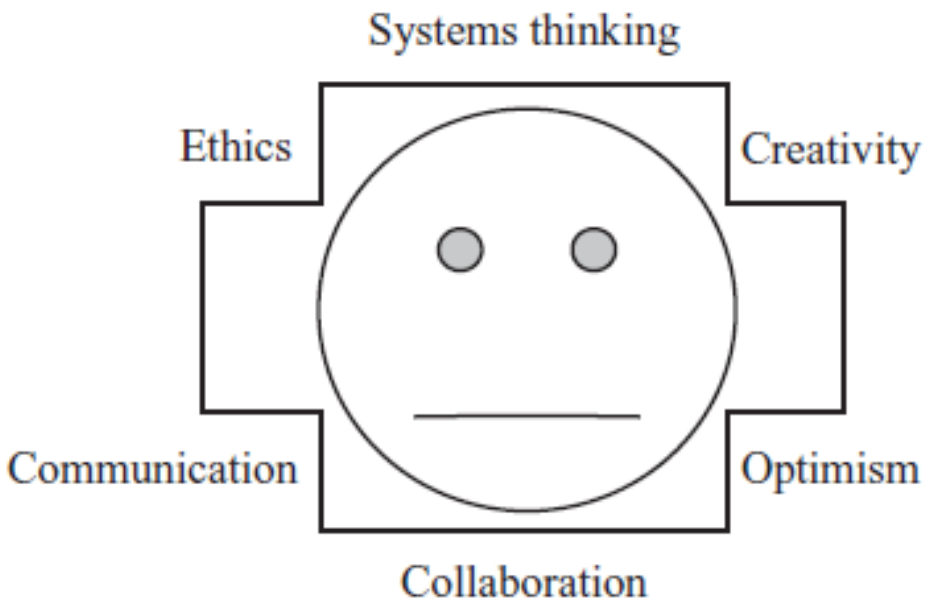
The term "**communication**" refers to the various forms of listening, speaking, writing, and non-verbal cues. Effective communication occurs when a whole message is sent completely received and understood.



The most important thing in communication is hearing what isn't said. Peter Drucker

Learning **Entrepreneurial Practice**

1. Persisting
2. Managing impulsivity
3. Listening with understanding and empathy
4. Thinking flexibly
5. Thinking about thinking
6. Striving for accuracy
7. Questioning and posing problems
8. Applying past knowledge to new situations
9. Thinking and communicating with clarity and precision
10. Gathering data through all senses
11. Creating, imagining, innovating
12. Responding with wonderment and awe
13. Taking responsible risks
14. Finding humor
15. Thinking interdependently
16. Remaining open to continuous learning



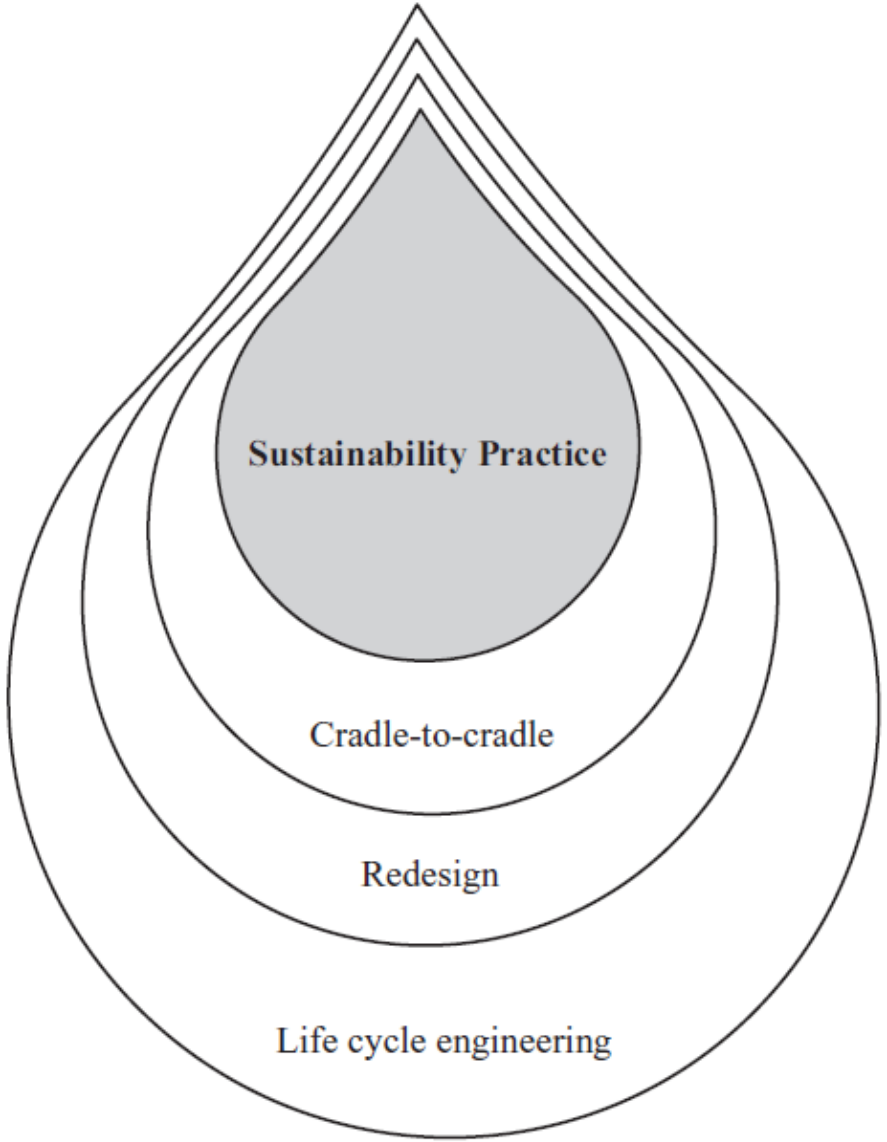
Learning **Safety in Design Practice**

Safety in design, in the sense perceived in the given quotation, is not a physical object, but a status in which danger is absent; it's not just about designing for simpler, easier use and long-term economic benefits through reduced production and maintenance costs, but for health and safety engaging standards and guidelines, change, remediation, and risk analysis.

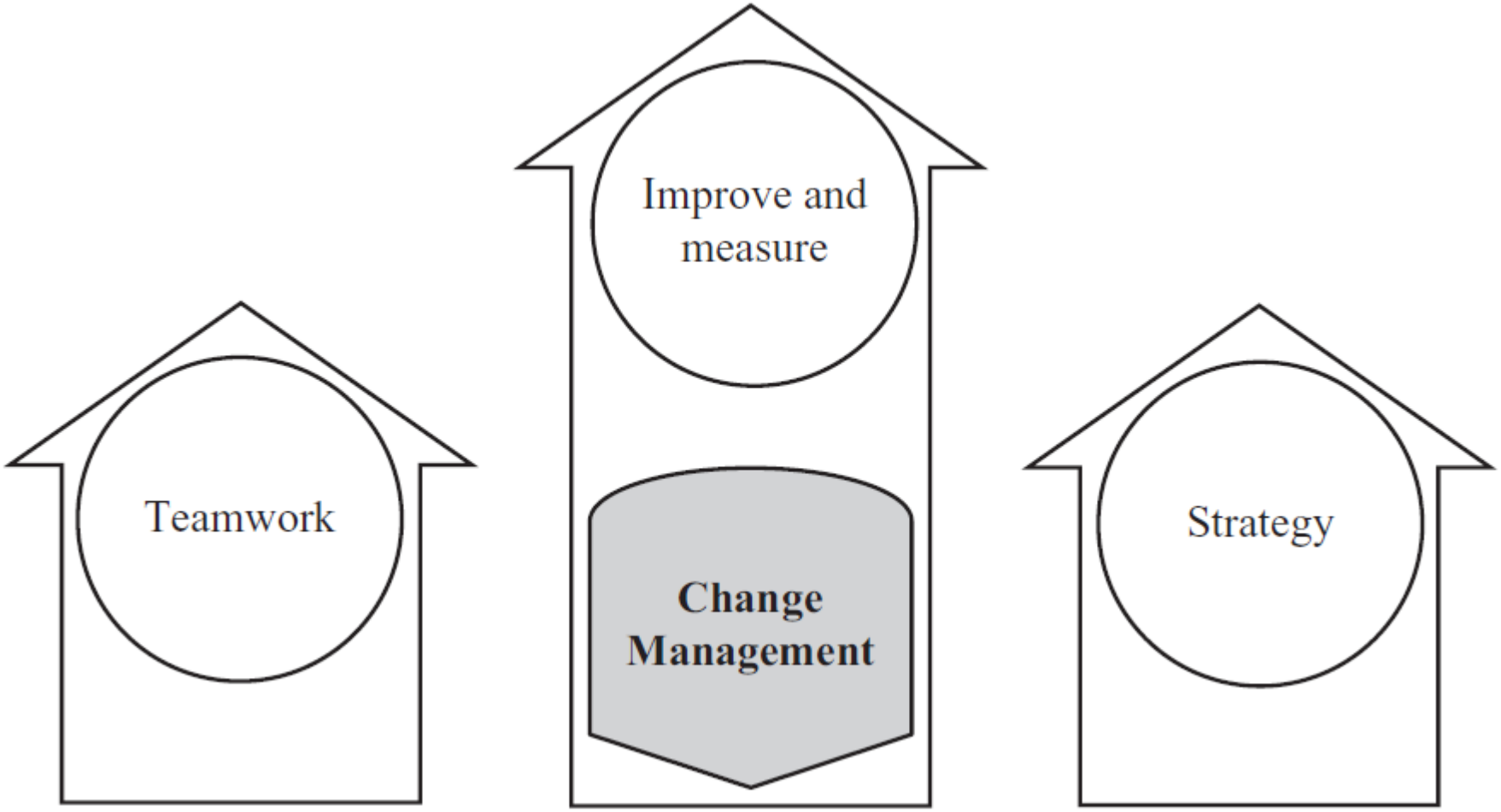
At the end of the day, the goals are simple: safety and security. **Jodi Rell**

Learning **Sustainability Practice**

Humanity has made significant progress since its origins; however, many individuals are still unable to lead fulfilling lives. Therefore, the United Nations has identified promoting good health and well-being as the third sustainable development goal (SDG) in its 2030 agenda.



Learning **Project Management Practice**

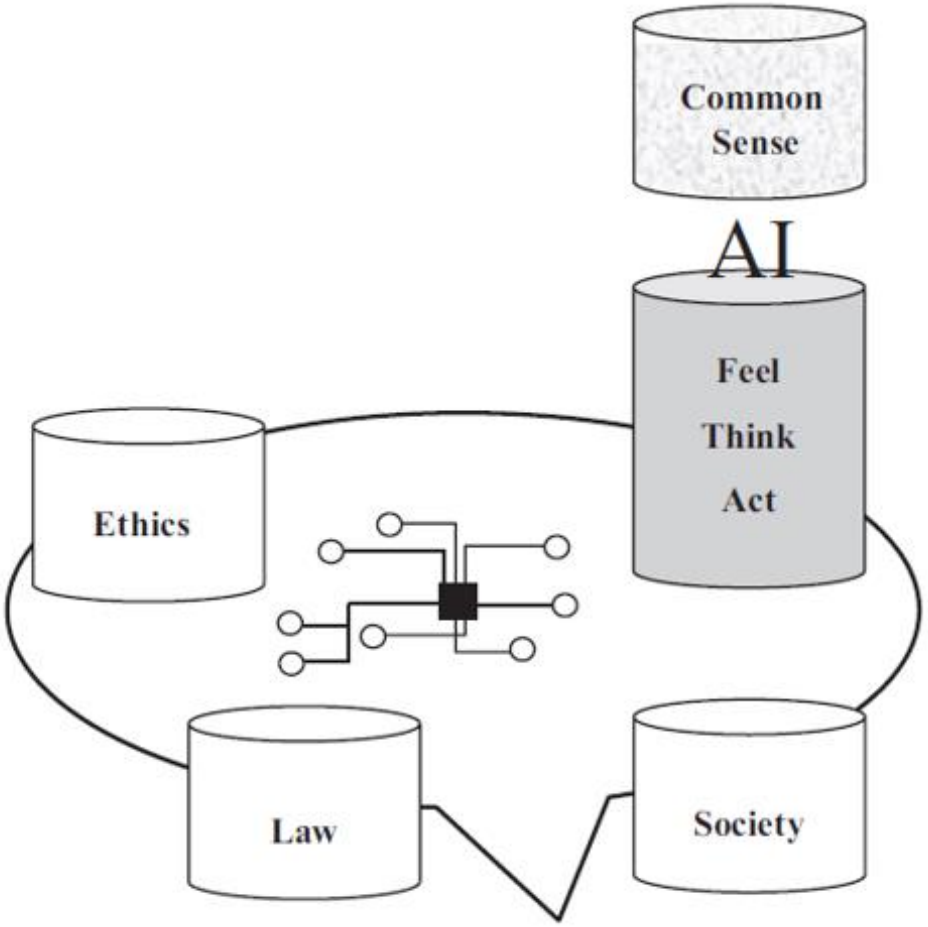


There is nothing permanent except **change**

Learning **AI Practice**

As a technologist, I see how AI and the Fourth Industrial Revolution will impact every aspect of people's lives. **Fei-Fei Li**

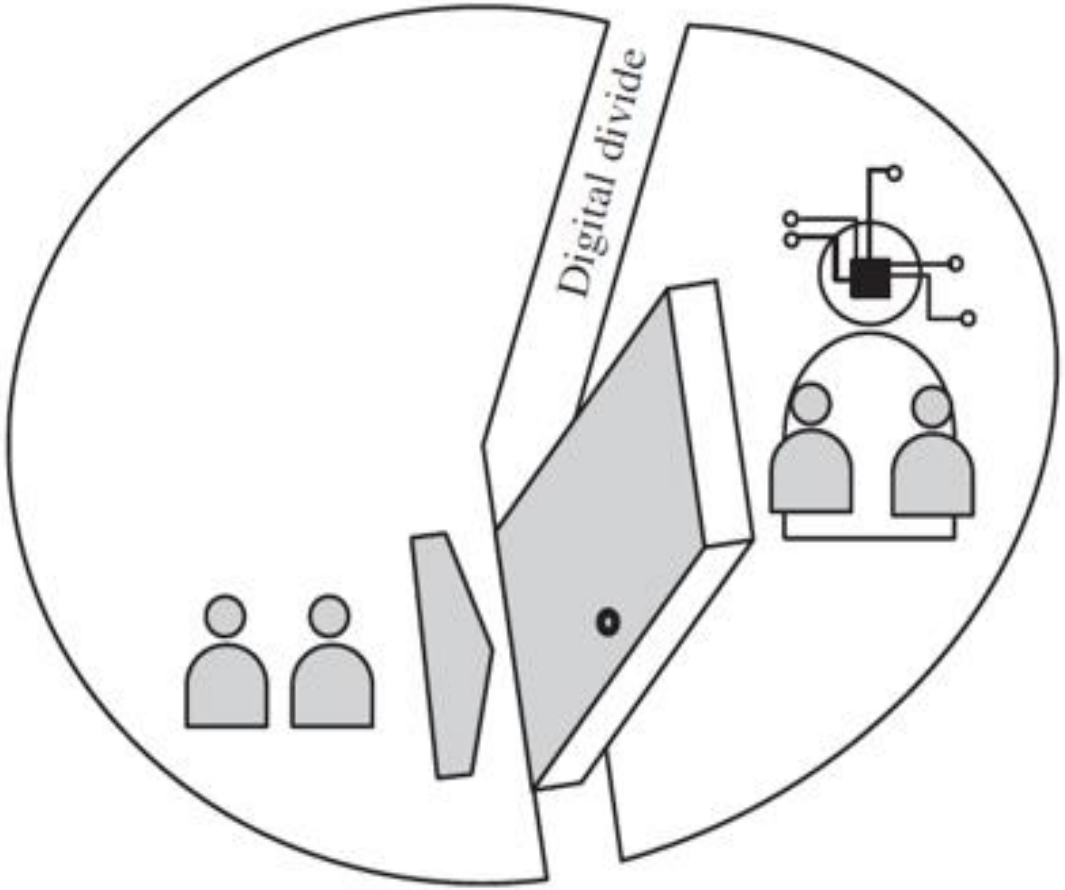
AI is not just about technology (**acting**), but about cognitive science (**feeling**) and computing (**thinking**) as well as about all social, ethical, and legal impacts on humanity.



Learning the **Changing Nature of Work**

If opportunity doesn't knock, build a door. **Milton Berle**

Today's doors to the digital world are changing mindsets and transforming the way people live, learn, and work significantly.



Research Case 1 (10 marks)

Equity in Engineering and Computing Practices

The evaluation of this open-ended task is determined by how well the content meets values, usefulness, writing skills, and relevance for its intended audience or purpose (use size **18; IEEE citation**).

Prepare a portfolio of FIVE PowerPoint slides as follows:

- **First Slide:** Create an e-poster that reflects on ethics and equity as essential skills (2 marks).

Investigate the following key areas, each on a slide. Develop an opinion(s) with proper in-text citation using at least two references.

- **Second Slide:** Cultivate equity in education by recognizing Anais Nin's words: "We don't see things as they are. We see them as we are" (2 marks).
- **Third Slide:** Equity in design and development (2 marks).
- **Fourth Slide:** Faculty/student equity experiences (2 marks).
- **Fifth Slide:** Equitable and inclusive teaching methods (2 marks).

- Submit the portfolio (all five slides) in one PDF document.
- Include your group number and the names of students at the top of the first slide.
- Create your own figures and images.
- The most effective content is that which is developed in a debate style.

Research Case 2 (10 marks)

Ethical Entrepreneurship Practices

The evaluation of this open-ended task is determined by how well the content meets values, usefulness, writing skills, and relevance for its intended audience or purpose (use size **18; IEEE citation**).

Prepare a portfolio of FIVE PowerPoint slides as follows:

- **First Slide:** Develop an **e-Poster** that reflects the entire portfolio (2 marks)

Investigate the following key areas, each on a slide. Develop an opinion(s) with proper in-text citation using at least two references.

- **Second Slide:** Value tension between profit and social responsibility (2 marks).
- **Third Slide:** Unique ethical challenges for startups (2 marks).
- **Fourth Slide:** The ethics of "fake it until you make it" (2 marks).
- **Fifth Slide:** Social entrepreneurship (2 marks).

- Submit the portfolio (all five slides) in one PDF document.
- Include your group number and the names of students at the top of the first slide.
- Create your own figures and images.
- The most effective content is that which is developed in a debate style.

Research Case 3 (10 marks)

Ethics of Care Practice

The evaluation of this open-ended task is determined by how well the content meets values, usefulness, writing skills, and relevance for its intended audience or purpose (use size **18; IEEE citation**).

Prepare a portfolio of FIVE PowerPoint slides as follows:

- **First Slide:** Develop an **e-Poster** that reflects the entire portfolio (2 marks).

Investigate the following key areas, each on a slide. Develop an opinion(s) with proper in-text citation using at least two references.

- **Second Slide:** Good care in engineering and computing (2 marks).
- **Third Slide:** Relational Ethics: Care beyond humans (2 marks).
- **Fourth Slide:** Moral judgments about care (2 marks).
- **Fifth Slide:** Bridging education and ethics of care practice (2 marks).

- Submit the portfolio (all five slides) in one PDF document.
- Include your group number and the names of students at the top of the first slide.
- Create your own figures and images.
- The most effective content is that which is developed in a debate style.

Research Case 4 (10 marks)

Responsible AI for Sustainable Development Goals (SDGs)

The evaluation of this open-ended task is determined by how well the content meets values, usefulness, writing skills, and relevance for its intended audience or purpose (use size **18; IEEE citation**).

Prepare a portfolio of FIVE PowerPoint slides as follows:

- **First Slide:** Develop an **e-Poster** that reflects the entire portfolio (2 marks).

Investigate the following key areas, each on a slide. Develop an opinion(s) with proper in-text citation using at least two references.

- **Second Slide:** Ethics, safety, and oversight foundation (2 marks).
- **Third Slide:** Building Verification & Validation(V&V) plan for AI in achieving SDGs (2 marks).
- **Fourth Slide:** Governance, certification thinking, and human oversight (2 marks).
- **Fifth Slide:** Assembling an insurance case outline (2 marks).

- Submit the portfolio (all five slides) in one PDF document.
- Include your group number and the names of students at the top of the first slide.
- Create your own figures and images.
- The most effective content is that which is developed in a debate style.

Project (30 marks)

Proof of Concept (PoC) + Proof of Value (PoV)

Themes: Relational Ethics, Responsibility, Values, Care, Empathy, Reflection, Reflexivity, PoC, STS.

Expected Skills: Hybrid (Personal + Interpersonal + Thinking + Communication + Technical)

In this project, an industrial partner entrusts a group of engineering and computing students, who are exploring the field of innovation, with the task of designing a "PoC + PoV" for a sustainable system that the company is interested in. The requirements include developing a market feasibility study, creating a pitch video, and preparing a grant application. **Therefore, each group should explore a real-world system of interest, where engineering, computing, and society intersect.**

Potential Research Questions:

What specific, real-world problem is this "PoC + PoV" solving, and for whom?

Who are all the potential stakeholders, including vulnerable user groups, and have their diverse needs and concerns been identified and addressed?

Have users been involved in the design process to gather feedback and ensure the solution aligns with their values?

What are the potential negative consequences or harms this "PoC + PoV" might cause to individuals, society, or the environment?

Feasibility Study (10 marks)

"Can we develop it?" and "Why is it worth building?"

The evaluation of this open-ended task is determined by how well the content meets values, usefulness, writing skills, and relevance for its intended audience or purpose (use size **18; IEEE citation**).

Develop a portfolio for the market feasibility study. It should be limited to FIVE slides. The following is a broad guideline; you may adapt it to suit your needs.

First Slide: Study title, along with an e-Poster that showcases the portfolio (2 marks).
Second Slide: Problem definition and the research questions (2 marks).
Third Slide: Literature/market review (at least five references) (2 marks).
Fourth Slide: Methodology and design (2 marks).
Fifth Slide: Solution's scope (2 mark).

Use the internet to research and understand the context of each of the above bullets.
Be realistic and artistic!

Pitch Educational Video (10 marks)

The best educational video should be short, engaging, inspirational, and well-structured using storytelling and strong audio-visuals to make a compelling case for the project.

A Persuasive Story: Structure the video with a clear beginning, middle, and end. Start with a captivating catch, such as a story or question, to seize attention immediately.

Problem and Solution: Clearly articulate the specific problem that the project addresses and explain its significance. Present your unique solution and demonstrate how it effectively solves the problem.

Target Audience: Tailor the message to the specific audience—whether they are investors, students, or clients—while addressing their pain points.

Proof and Integrity: Build trust by including evidence such as user testimonials, data, research findings, or a demonstration of the project idea in action.

Action: Conclude with a direct and simple instruction for the audience regarding what to do next,

Concise: Keep the video short, about **TWO** minutes, focusing on the most crucial information to maintain engagement and avoid cognitive overload.

Grant Application (10 marks)

The best grant applications are clear, compelling, well-researched, and carefully aligned with the funder's specific goals and guidelines.

Strong Alignment: The application clearly demonstrates how the project's goals and outcomes align with the funding organization's mission, values, and priorities.

Need Statement: It effectively articulates the need or problem that the project will address, using reliable data, statistics, and impactful real-world examples.

Measurable Objectives: The objectives are "specific, measurable, achievable, relevant, and time-bound," providing clear indicators of success and enabling the funder to trace progress.

Plan and Methodology: The proposal outlines a realistic and detailed action plan, including specific activities, a clear timeline, and a robust methodology. This explains how the objectives will be achieved and why this approach is the most effective.

The grant application form is available in the **project** folder. **Be reflective!**